

Measurement Changes Phenomenon Classroom Run Kit

TOK Cognitive Science Activity Bank • Natural Sciences • 25-35 min

Category	Natural Sciences	Time	25-35 min
Difficulty	Easy	ID	measurement-changes-phenomenon

Big idea: Measurement does not merely capture phenomena; it operationalizes them.

One-Minute Teacher Brief

Students compare different ways of measuring the same phenomenon and see how instruments shape the result.

Student Prompt

Inspect Measurement Changes Phenomenon Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Measurement Changes Phenomenon Stimulus A	Student-facing stimulus 1: Measure classroom attention by eyes on speaker, notes taken, response accuracy, or self-reported focus. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Measurement Changes Phenomenon Stimulus B	Student-facing stimulus 2: Measure noise by peak loudness, average loudness, teacher annoyance, or number of interruptions. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Measurement Changes Phenomenon Reveal / Twist	Student-facing stimulus 3: Measure noise by peak loudness, average loudness, teacher annoyance, or number of interruptions. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Measurement does not merely capture phenomena; it operationalizes them.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Measurement Changes Phenomenon Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Measurement Changes Phenomenon Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.

#	Question for students	Teacher key / cue
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Does measurement reveal or define what is being studied?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how measurement does not merely capture phenomena; it operationalizes them.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Does measurement reveal or define what is being studied?

Setup Checklist

- Choose a phenomenon that can be measured quickly in several plausible ways.
- Make each group use a different operational definition before comparing results.

Example Stimuli or Materials

- Measure classroom attention by eyes on speaker, notes taken, response accuracy, or self-reported focus.
- Measure noise by peak loudness, average loudness, teacher annoyance, or number of interruptions.

Resources To Use

- Resource pack: <resources/measurement-changes-phenomenon/index.html>
- Card deck CSV: <resources/measurement-changes-phenomenon/cards.csv>
- Visual prompt: <resources/measurement-changes-phenomenon/visual-prompt.svg>
- Measurement design card: concept, operational definition, tool, weakness.
- Comparison table for results from different measures.

Run Sequence

1. Choose one phenomenon and ask groups how it could be measured.
2. Assign different tools or operational definitions to different groups.
3. Groups collect quick data using their assigned method.

4. Compare results and explain why the same phenomenon produced different data.
5. Debrief the relationship between measurement, concepts, and evidence.

Debrief Moves

- Knowledge question: Does measurement reveal a phenomenon or partly define it?
- When do different measures count as evidence for the same thing?
- How do instruments extend and constrain scientific knowledge?

Teacher Quick Script

- Before we ask what the data says, ask what the data is a measure of.
- Did your method measure the phenomenon or one convenient version of it?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...